

Backlog Buddy

Demonstration question paper generated from available previous papers - for practice only

B.Tech. II Semester Examination, 2026

Regulation: R20 | University: JNTUK

Database Management Systems (21CS42)

Time: 3 hours Max. Marks: 60

Answer one question from each unit. All questions carry equal marks.

UNIT - 1 [Introduction & Relational Model]

- a) Explain the three schema architecture of a DBMS with a diagram.
- b) Define the different types of keys in the relational model with examples.

UNIT - 2 [SQL & Procedural SQL]

- a) Explain the different types of joins in SQL with examples.
- b) Write a trigger and a stored procedure for the given schema.

UNIT - 3 [Normalization & Functional Dependencies]

- a) Compute the closure of the given functional dependencies and find all candidate keys.
- b) Normalize the given relation up to BCNF with justification.

UNIT - 4 [Transactions, Concurrency & Recovery]

- a) Explain the ACID properties of a transaction.
- b) What is deadlock? Explain deadlock prevention and detection techniques.

UNIT - 5 [Storage, Indexing & Advanced Databases]

- a) Explain B+ tree indexing with an example.
- b) Write short notes on data warehousing and OLAP.